Parallels between 'road' and 'research data' infrastructures

Data-intensive research depends on data and data services, such as databases, software and tools, and standards. These are often made available to end-users through research infrastructure.

As is common for many types of infrastructure, the existence of research infrastructure is often taken for granted by their users. Their importance is only noticed when they are (temporarily) unavailable.

As research infrastructures are typically funded from public money, it is essential that policy makers, research funders, experts reviewing funding proposals, and possibly even end-users know about the daily tasks of service providers. We have noticed that this is not often the case. We set out to improve this situation by comparing components of data infrastructure with more common road infrastructure.

Find more in our policy brief:

Components

Data Infrastructure	Road Infrastructure
Data service	Road
User of a data service / Researcher	Car driver
Research project	Car
Cutting edge research project	Expensive car
Project	Trip, journey
Service operator	Road construction company
Flexible data service	Road suitable for cargo and people
Hack something together yourself	Driving your own Jeep off-road
Using an existing service	Driving on a motorway
Services that the user pays for	Toll road
Overused service	Traffic Jam
Services that can be used by anyone	Public roads
Interoperable services	Road intersections
Non-interoperable services	Crossing roads without intersections
Efficiency of a thought-out service	Speed of a Ferrari on an asphalt road
Efficiency of figuring things out	Speed of a Jeep on rough terrain
What can be realistically achieved in a scientific project	How far you can travel in a day
ELIXIR's visibility in biology	Public knowledge about the companies building roads
ELIXIR's visibility in bioinformatics	Knowledge in the government about the companies building roads
Infrastructure investing in services based on community needs	Government prioritising road investments based on transport needs
e-Infrastructure: infrastructure under the infrastructure.	An asphalt factory. No road construction without them
Assistance identifying the optimal combination of services that could be used in a project.	Navigation system

Policy considerations

Data Infrastructure	Road Infrastructure
Thinking data infrastructure is too expensive (or lacking sufficient accommodation of one's specific needs) and preferring to build everything yourself to exact specifications; underestimating the value of services and also the real cost of building everything yourself (excluding e.g. energy bills, and the fact that the postdoc is not productive for work while making weekly backups)	Not being willing to pay road taxes or to pay another centralised tax system (or complaining that you have to walk the last bit from the parking lot), underestimating the value provided (to its users) by the road infrastructure and underestimating the cost incurred to get exactly from A->B without roads (i.e. using an all-terrain car), e.g. forgetting that there won't be any gas stations along the way either.
Not knowing whether the infrastructure will live long enough to serve your entire project (and its successor), and therefore being coerced into build it yourself	Not knowing whether the roads will be maintained for 10 years, and therefore better off buying a Jeep than either a Ferrari or a Volkswagen
Asking a research infrastructure provider what scientific breakthroughs the infrastructure will be making. An RI does not make breakthroughs nor predict what they will be.	Asking a road construction & maintenance company how much faster the distance will be travelled. The road construction company cannot predict what cars people will use and what they will carry.
Making the requirement that a funding proposal for "infrastructure" includes scientific breakthroughs to happen in a fixed timespan	Asking a road company to deploy a combination of a very fast asphalt depositing machine, a Ferrari, and a system to clean up the road immediately afterwards
Asking top scientists to review proposals for research infrastructure funding. The services that would be offered by the infrastructure would be "competing" with what they (and only similar top scientists) can get done in their own labs	To ask those who have expensive Jeeps whether they approve the construction of a public road. They and others with off-road cars do not see the need for roads, they can get along just fine without them

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